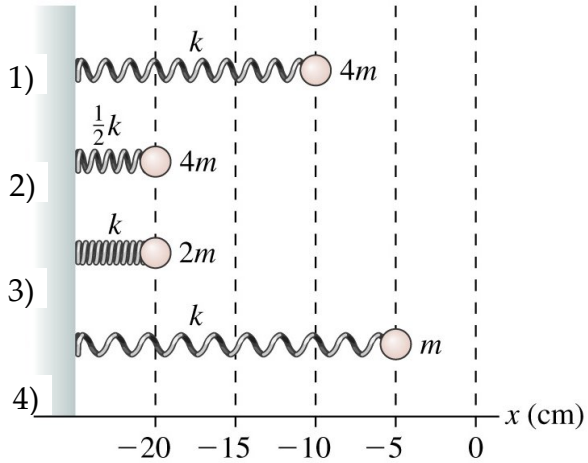
Section A: Multiple Choice Questions

1) One person swings on a swing and finds that the period is 3.0 s. Then a second person of equal mass joins him. With two people swinging, the period is

A. 1.5 s.
B. <3.0 s but not necessarily 1.5 s.
C. 3.0 s.
D. >3.0 s but not necessarily 6.0 s.
E. 6.0 s.

2) Four springs have been compressed from their equilibrium position at x = 0 cm. When released, they will start to oscillate. Rank in order, from highest to lowest, the maximum speeds of the oscillators.



A. $v_3 > v_2 > v_1 = v_4$
B. $v_3 > v_2 > v_1 > v_4$
C. $v_4 > v_1 > v_2 > v_3$
D. $v_1 = v_4 > v_2 > v_3$
E. $v_2 > v_3 > v_1 = v_4$

3) Rank the waves represented by the following 3 functions according to their **period**, from the largest to the smallest.

$y1 = 2 \sin(3x + 10 t + \pi)$

$y2 = 8 \cos(6x + 19 t)$

$y3 = 7 \sin(4x + 24 t + 6)$

In the 5 options listed, T1 refers to the period of y1, etc.

Choose the correct option below:

A. T3>T2>T1

B. T1>T2>T3

C. T1>T3>T2

D. T2>T3>T1

E. T1>T3>T2

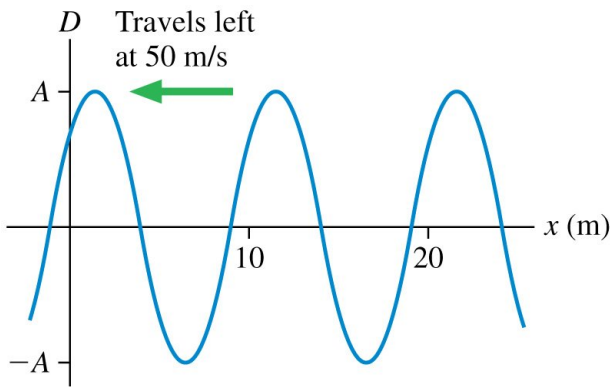
4) Which of the following actions would make a pulse travel **faster** down a stretched string?

A. Move your hand up and down **more quickly** as you generate the pulse.
B. Move your hand up and down **a larger distance** as you generate the pulse.
C. Use a **heavier** string of the same length, under the same tension.
D. Use a **lighter** string of the same length, under the same tension.
E. Use a **longer** string of the same thickness, density and tension.

5) If the string does not stretch, by what factor would you have to multiply the tension in a taut string so as to double the wave speed?

A. 0.25
B. 4
C. 2
D. 0.5
E. You could not change the speed by changing the tension.

6) What is the frequency of this traveling wave (see figure)?



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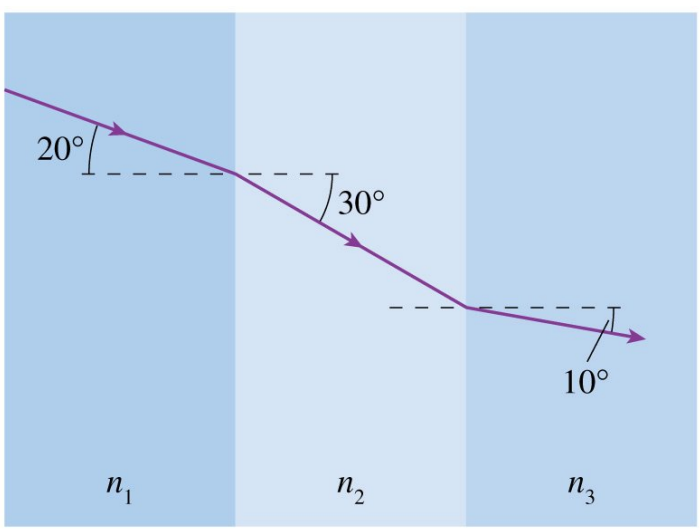
Y

A. 10 Hz
B. 5 Hz
C. 2 Hz
D. 0.2 Hz
E. 0.1 Hz

7) A light ray travels from medium 1 to medium 2, then to medium 3 as shown. The dashed lines are normal to the interfaces.



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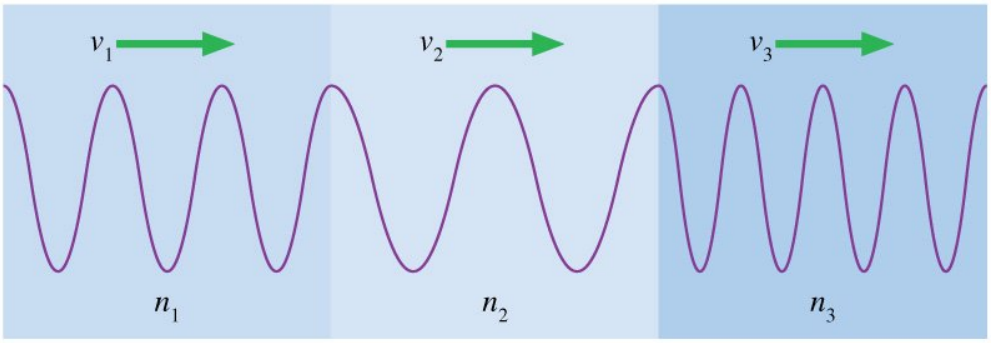
For these media, which relationship between indices of refraction is correct?

A. $n_3 < n_1$
B. $n_3 = n_1$
C. $n_3 > n_1$
D. We can't compare n_3 to n_1 without knowing n_2

8) A light wave travels through three transparent materials of equal thickness. Rank in order, from largest to smallest, the indices of refraction n_1, n_2, and n_3.



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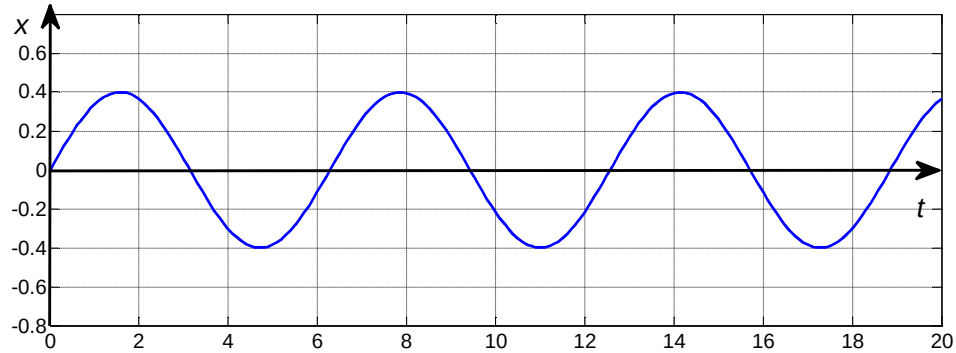
A. $n_2 > n_1 > n_3$
B. $n_3 > n_1 > n_2$
C. $n_1 > n_2 > n_3$
D. $n_3 > n_2 > n_1$
E. $n_2 = n_1 = n_3$

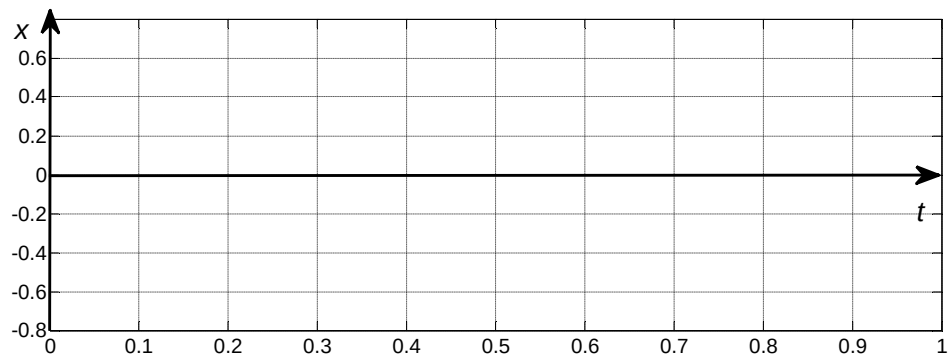
Section B: Long-Answer Questions

1) The graph below shows the position versus time for a mass oscillating on a spring, in some suitable units.

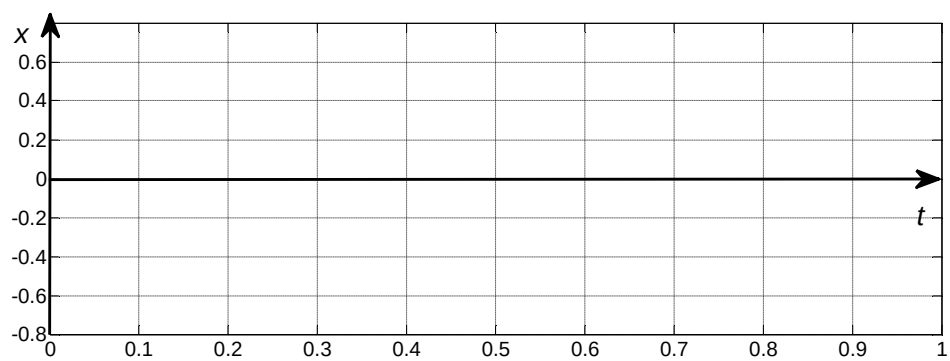


On the axes below sketch the position-versus-time plots for the following situations which differ in some respects from the original one. **Note: The changes described in each part refer back to the <u>original oscillation</u>, not to the oscillation of the previous part of the question**. Use the same horizontal and vertical scales as in the original graph.

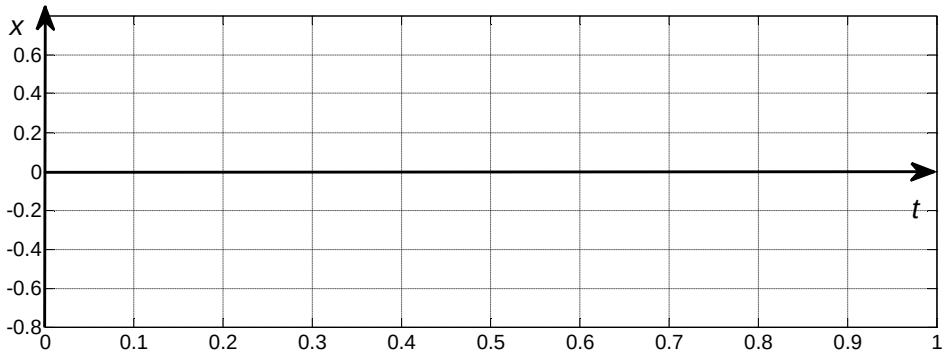
a) The amplitude and frequency are doubled.



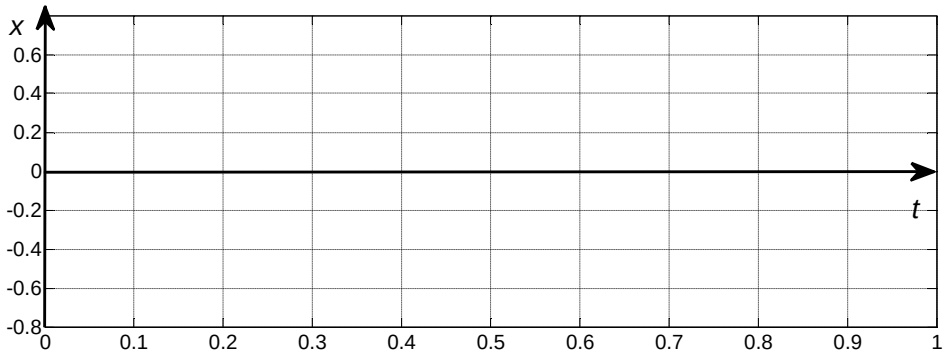
b) The amplitude is halved and the mass is quadrupled.



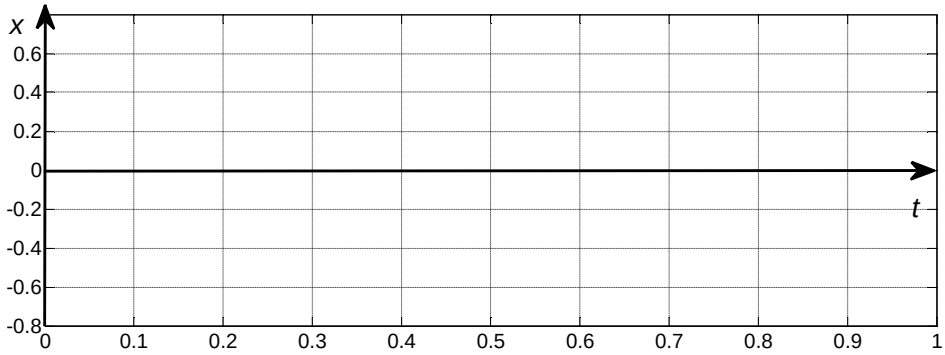
c) The phase constant is increased by $\pi/2$ rad.



d) The maximum speed is doubled while the amplitude remains constant.



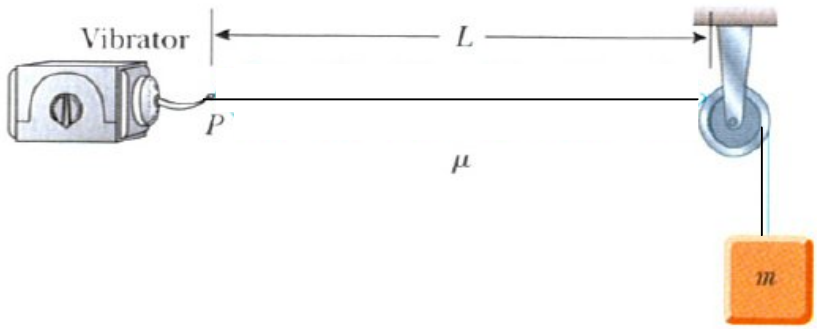
e) The kinetic energy is quadrupled while the maximum speed remains constant.



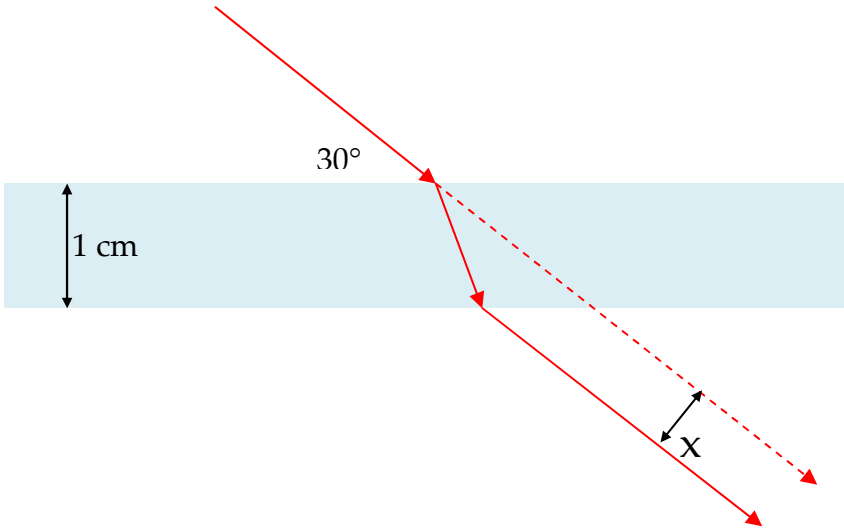
2) Damping is negligible for a 0.100 kg object hanging from a light spring with spring constant 6.40 N/m. A sinusoidal force with amplitude of 1.00 N drives the system. At what angular frequencies will the force make the object vibrate with an amplitude of 0.250 m?

Hint: $A = \dfrac{F_0 / m}{\sqrt{\left(\omega^2 - \omega_0^2\right)^2 + \left(\dfrac{b}{2m}\right)^2}}$

3) In the arrangement shown an object is hung from a string (with a linear mass density of 0.00200 kg/m) that passes over a light pulley. The string is connected to a vibrator of frequency f=350Hz. The length of the string between point P and the pulley is L=2.00 m. What is the largest mass, *m*, for which a standing wave can be observed?



4) Find the lateral displacement X (see figure) of the light beam passing through a planar transparent plate of thickness 1 cm and index of refraction 1.4, surrounded by air.



Keys

Section A: Multiple-Choice Questions

1. C
2. A
3. B
4. D
5. B
6. B
7. C
8. B

Section B: Long-Answer Questions

2. ω = either 10.2 or 4.90 rad/s
3. Largest mass = 400 kg
4. x = 0.47 cm